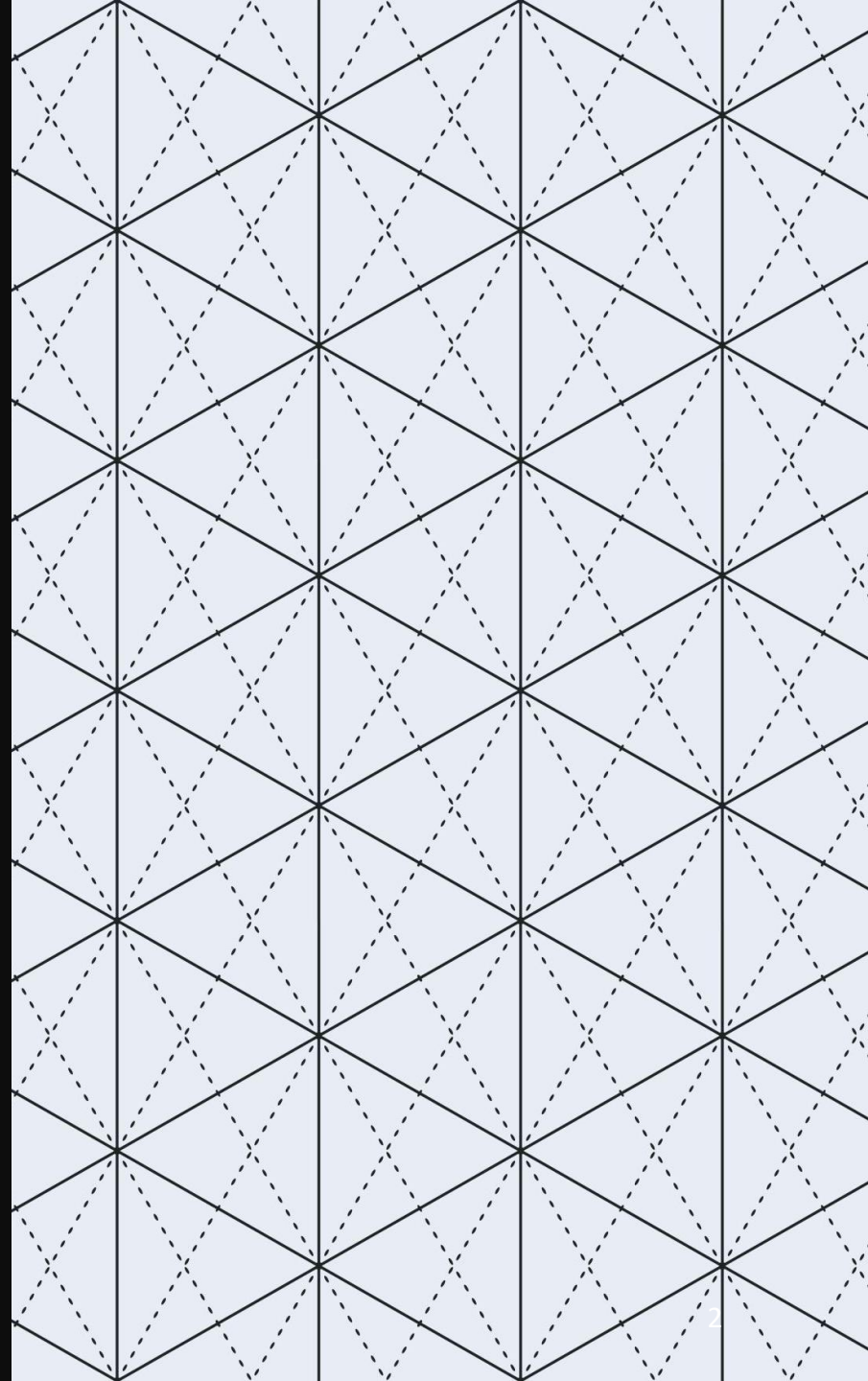


Direct and Inverse Proportion

Grade 10 Extended Maths
AY 2025 – 2026

Outline

- Direct Proportion
 - Inverse Proportion
-



Learning Objectives



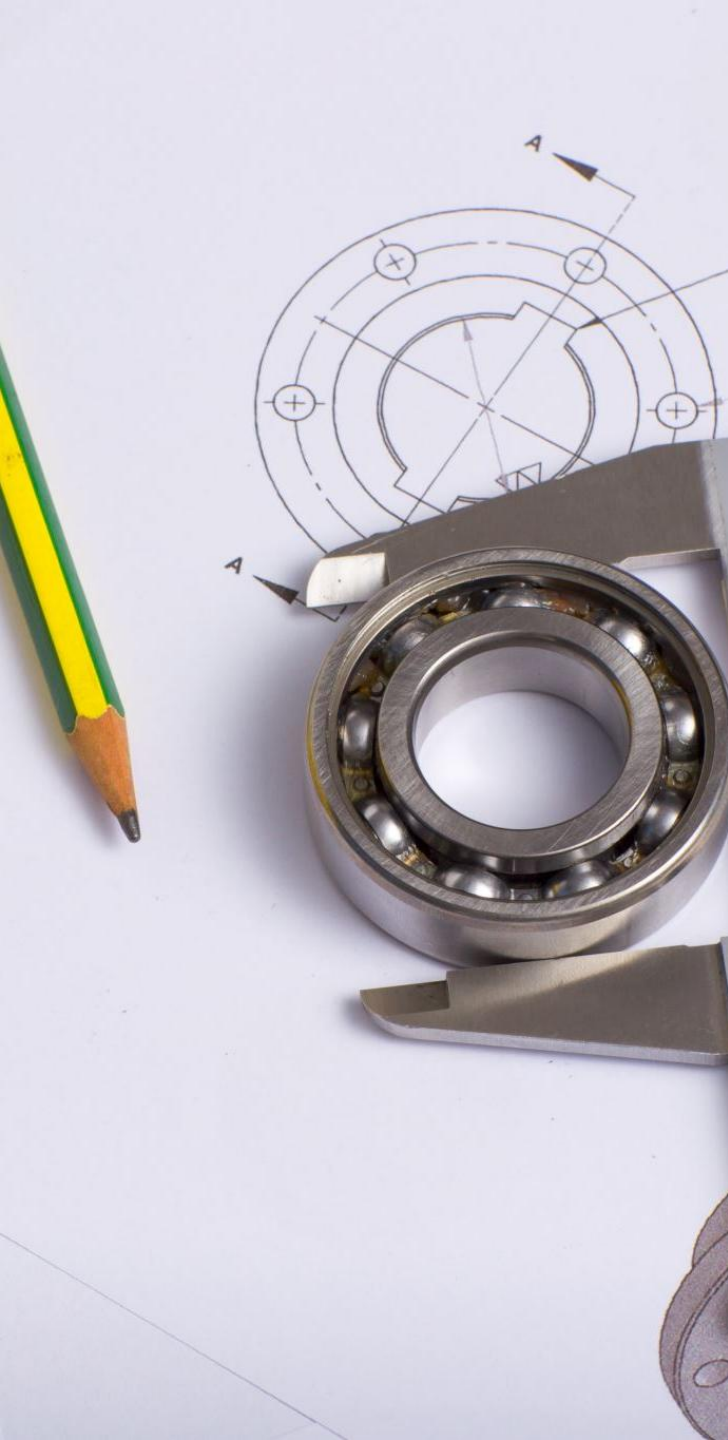
Express direct proportion in algebraic terms



Express inverse proportion in algebraic terms



Use the algebraic expression of a direct or inverse proportion to find unknown quantities.



Applications

- Daily life – kinematics, groceries
- Mathematical modelling
- Others?

Learning Objectives

- Express direct proportion in algebraic terms
- Express inverse proportion in algebraic terms
- Use the algebraic expression of a direct or inverse proportion to find unknown quantities.

The background features handwritten mathematical notes on a chalkboard. At the top left, the equation $y = g(x)$ is written. Below it, the words "Secant Lines" are written. To the right, the limit definition of a derivative is shown: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$. Below this, the same definition is repeated with a specific function: $f(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$. This is followed by the expansion of the numerator: $= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$. Finally, the expression is simplified to: $= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$. Other faint notes include "Tangent Line" and "x+h".

Case Study 1

In a supermarket, 6 kg of rice cost \$24.
What would be the cost of 13 kg of rice?

Case Study 1 (discussion)

In a supermarket, 6 kg of rice cost \$24.

What would be the cost of 13 kg of rice?

6 kg of rice $\Leftrightarrow$ \$24

1 kg of rice $\Leftrightarrow \frac{\$24}{6} = \4

13 kg of rice $\Leftrightarrow 13 \times \$4 = \$52$

What about the cost of 18 kg of rice? 47 kg of rice?

What would be the maximum amount of rice can be purchased with \$60? With \$84?

How can we solve any related problems easier?

Formalising the concept – discussion

- What do you notice about the relationship between the amount of rice (R) and its cost (C)?

Direct Proportion - Summary

- When two quantities are related in such a way that the increment (or decrement) of one quantity increases (or decreases) the other quantity **at the same rate**, we call the two quantities to be directly proportional.
- When two quantities are **directly proportional** to each other they are linked by a constant multiplier (known as the **Constant of Proportionality**). This means that the **ratio** of the two variables will always be this constant.

Direct Proportion - Summary

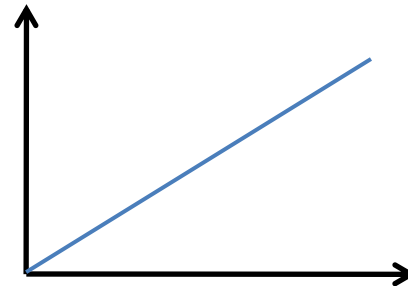
- In general, if “**y is directly proportional to x**” then we have...

$$y \propto x \leftrightarrow y = kx \leftrightarrow k = \frac{y}{x}$$

Proportionality symbol

Constant of Proportionality

Note: A graph of two variables that are **directly proportional** to each other will be a **straight-line graph through the origin** ($y = kx$)!



Direct Proportion - Summary

- Some examples can be found on the table below whereas Quantity 1 is directly proportional to Quantity 2.

Quantity 1	Quantity 2	Expression
y	x	$y \propto x \leftrightarrow y = kx$
y	x^2	$y \propto x^2 \leftrightarrow y = kx^2$
y	$\sqrt{x}$	$y \propto \sqrt{x} \leftrightarrow y = k\sqrt{x}$
y	$\frac{1}{x}$	$y \propto \frac{1}{x} \leftrightarrow y = \frac{k}{x}$

Direct Proportion - Examples

Example 1

Two quantities y and x are directly proportional to each other.

When $x = 4$, $y = 10$.

- (a) Find a formula for y in terms of x .
- (b) Find the value of y when $x = 6$.
- (c) Find the value of x when $y = 8$.

Direct Proportion - Examples

Example 2

The table below shows a set of matching p and q values, where p is directly proportional to q . Find a formula for p in terms of q , and then work out the missing values in the table.

p	1	3		30
q		12	18	

Direct Proportion - Examples

Example 3

y is directly proportional to x^2 . Given that $y = 144$ when $x = 4$, find

- a) an expression for y in terms of x ,
- b) the value(s) of x when $y = 2025$,
- c) the percentage increase or decrease in the value of y when x is doubled.

Case Study 2

If it takes 5 workers 150 days to build 1 house,

- a) how long would it take for 10 workers to build 1 house? How about 12 workers?
- b) how many workers will be needed to build 1 house in 125 days? How about 100 days?

Case Study 2 (discussion)

If it takes 5 workers 150 days to build 1 house,

a) how long would it take for 10 workers to build 1 house?

5 workers $\Leftrightarrow$ 150 days

1 worker $\Leftrightarrow 150 \times 5 = 750$ days

10 workers $\Leftrightarrow \frac{150 \times 5}{10} = 75$ days

Case Study 2 (discussion)

If it takes 5 workers 150 days to build 1 house,

b) how long would it take for 12 workers to build 1 house?

5 workers $\Leftrightarrow$ 150 days

1 worker $\Leftrightarrow 150 \times 5 = 750$ days

12 workers $\Leftrightarrow \frac{150 \times 5}{12} = 62.5$ days

Case Study 2 (discussion)

If it takes 5 workers 150 days to build 1 house,

c) how many workers will be needed to build 1 house in 125 days?

5 workers $\Leftrightarrow$ 150 days

1 worker $\Leftrightarrow 150 \times 5 = 750$ days

12 workers $\Leftrightarrow \frac{150 \times 5}{12} = 62.5$ days

Formalising the concept – discussion

- What do you notice about the relationship between the number of workers (W) and number of days (D)?
- Similar to the discussion on Direct Proportion, form a formula to connect the number of workers (W) and number of days (D).

Inverse Proportion - Summary

- When two quantities are related in such a way that the **increase** of one quantity causes a **decrease** in the other quantity at a proportional rate, we call the two quantities to be **inversely proportional**.
- When two quantities are **inversely proportional** to each other, the **product** of the two quantities will be a constant.

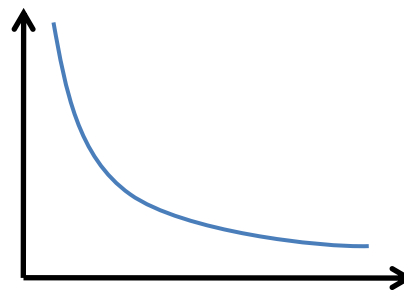
Inverse Proportion - Summary

- In general, if “**y is inversely proportional to x**” then we have...

$$y \propto \frac{1}{x} \leftrightarrow y = \frac{k}{x} \leftrightarrow xy = k$$

Product is constant

Note: A graph of inverse proportion will look like this...



Inverse Proportion

Example 4

Two quantities x and y are such that y is inversely proportional to x . When $x = 20$, $y = 6$.

- (a) Find a formula connecting x and y .
- (b) Work out the value of y when $x = 40$.
- (c) Work out the value of x when $y = 2$.

Inverse Proportion

Example 5

In the table below, f is inversely proportional to g . Find a formula for f in terms of g and then use it to find the missing values in the table.

f		10	40
g	45		9

Inverse Proportion

Example 6

p is inversely proportional to the cube of v .

- (a) It is known that $p = 4$ for a particular value of v . What is the value of p when v is doubled?
- (b) Given that $p = 1$ and $v = 2$, write down a formula connecting p and v .
- (c) Hence find the value of v when $p = 125$.

Reflection/Journal

Write your reflection/journal by answering the following question

- 1) What does a relationship of direct proportionality tell us?
- 2) What does a relationship of inverse proportionality tell us?
- 3) How is proportionality different from a linear equation?

Textbook Exercises

Direct Proportion

Textbook Exercises 21.1 No. 2, 4, 5, 7, 8

Inverse Proportion

Textbook Exercises 21.2 No. 1, 2, 4, 5, 7